

Data communications (Lab 5): Unipolar NRZ Encoding

Objective: To generate and analyze a Unipolar NRZ (Non-Return-to-Zero) encoded signal using MATLAB from a given binary data sequence.

Theory

Unipolar NRZ is a line coding technique used in digital communication to represent binary data using a single positive voltage level and zero voltage.

In Unipolar NRZ encoding:

- Binary 1 $\rightarrow$ positive voltage level
- Binary 0 $\rightarrow$ zero voltage level

The signal does not return to zero during the bit interval, which is why it is called Non-Return-to-Zero (NRZ).

Thus:

- 1 is represented by a constant high level
- 0 is represented by a constant zero level

MATLAB Code

```
clear;
clc;

b = [1 0 1 1 1 1 0];
n = length(b);
t = 0:0.01:n;

for i = 1:n
    bw(i*100:(i+1)*100) = b(i);
end

bw = bw(100:end);

subplot(1,1,1)
plot(t,bw)
grid on;
axis([0 n -2 +2])
title('Unipolar NRZ Encoded Signal')
xlabel('Time')
ylabel('Amplitude')
```

Code Explanation

1. Initialization

```
clc;  
clear;  
close all;
```

- `clc` → Clears command window
- `clear` → Removes all variables

2. Define Binary Input

```
b = [1 0 1 1 1 1 0];  
n = length(b);
```

- `b` contains the binary input sequence
- `n` gives the total number of bits

3. Create Time Vector

```
t = 0:0.01:n;
```

- Creates the time axis from 0 to `n`
- Step size 0.01 gives a smooth plot

4. Generate Unipolar NRZ Waveform

```
for i = 1:n  
    bw(i*100:(i+1)*100) = b(i);  
end
```

- Each bit is expanded over 100 samples
- If the bit is 1, the waveform stays at 1
- If the bit is 0, the waveform stays at 0
- This creates a rectangular digital waveform.

5. Adjust Waveform Length

```
bw = bw(100:end);
```

- Removes extra initial samples
- Matches the length of `bw` with the time vector `t`

6. Plotting

```
subplot(1,1,1)  
plot(t,bw)  
grid on;  
axis([0 n -2 +2])
```

- Plots the encoded waveform
- `grid on` adds grid lines
- `axis([0 n -2 +2])` sets the x and y axis limits

Result:

The MATLAB program produces the Unipolar NRZ encoded waveform for the given binary sequence.

Observations:

- For binary 1, the signal remains at a positive level
- For binary 0, the signal remains at zero
- The signal does not return to zero during the bit duration

Conclusion: The experiment successfully demonstrated Unipolar NRZ encoding using MATLAB. The binary input sequence was converted into a digital waveform in which 1 is represented by a high level and 0 is represented by zero level. The output verified the basic principle of Unipolar NRZ line coding.

